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AI Action Plan
Attn: Faisal D'Souza
NITRO NCO
2415 Eisenhower Avenue
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March 14, 2025

Dear Mr. D'Souza

Ernst & Young **LLP** (EY US) is pleased to submit comments¹ in response to the White House Office of Science and Technology Policy (OSTP) and Networking and Information Technology Research and Development (NITRO) National Coordination Office, which seeks input on the development of an artificial intelligence (AI) action plan.

The opportunities presented by the widespread deployment of AI technologies could potentially lead to scientific breakthroughs, increase the efficiency of business operations, conserve resources for complex and creative thinking, and improve livelihoods. To achieve these outcomes, AI must be deployed in a way that manages risks to all stakeholders and instills confidence in the processes and outcomes fueled by AI. Our submission provides insights on AI risks and governance, cybersecurity, and explainability; three topics that, if addressed successfully, will accelerate adoption of AI and the realization of the incredible potential that AI technology holds for the United States of America.

We have a unique perspective in the field of AI and machine learning given our network's global, cross-sector footprint and our experience developing and deploying AI systems. Our team of dedicated data scientists advises entities across every sector of the economy on adopting a holistic view of automation, process and service improvement. We help organizations craft, deploy and evaluate AI tools to responsibly accomplish their digital transformation goals.

EY US is the US member firm of the global network of EY member firms that provides advisory, assurance, tax and transaction services to entities worldwide. Together our global network comprises approximately 54,000 of the more than 365,000 EY professionals worldwide and serves many companies across a wide range of industries in the US.

AI risks and governance

Risks introduced by AI systems

Exponential growth in the adoption of AI introduces significant risks that must be managed. These risks stem from various factors, including inaccurate data sets, lack of transparency in certain AI models, and insufficient human intervention. These factors can create risks to accuracy, impartiality and reliability. Other risks that are gaining attention include value chain dependencies and "shadow AI" where employees use non-approved AI for business purposes, potentially undermining data security, and cybersecurity. Understanding who will use these tools, when and how they will be deployed, and their specific purposes is essential in qualifying these risks, as well as implementation of appropriate mitigation techniques.

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Although risks will depend on various factors, including use cases and AI models, key risks associated with AI adoption include:

- Inaccuracy from predictive AI: Predictive models trained on historical data may fail to accurately forecast future events, particularly under extreme conditions.
- Lack of human oversight and intervention capability: The absence of human oversight in AI systems can lead to adverse outcomes, especially when using AI to expedite operations without proper checks.
- Data privacy violations: AI systems can raise significant privacy concerns, including data persistence, repurposing, security of intellectual property, and spillover impacting both personal data and corporate data.
- Partiality: Algorithmic partiality can reflect human partiality, leading to systematic errors in outcomes.
- Value-chain dependencies:
 - With AI systems increasingly built on top of large AI foundation models, there are risks associated with:
 - Service bottlenecks: a single point of failure at the AI model vendor risking a systemic failure to all downstream services that depend on the same AI model
 - Inability of downstream AI systems to be able to switch to a different foundation model provider without significant re-tooling
 - Power grid and data network load capacity limits
- Operations and infrastructure challenges:
 - Insufficient knowledge about AI among businesses can expose them to risk due to:
 - Inability to assess if they are buying the right AI for the job
 - Or
 - Inability to assess if the AI system is operating as it should be
 - Tech infrastructure and energy capacity limitations
 - Inadequate worker resourcing

Data challenges

AI models rely on high-quality, secure and structured data to operate effectively, thus ensuring data integrity and security is essential for AI trustworthiness. The need for robust data governance and scalable infrastructure is becoming increasingly critical as AI adoption increases. A primary challenge lies around data fragmentation, as data is often stored and managed across multiple sources, formats and platforms. Without proper data governance, AI systems may be trained on inconsistent or incomplete data sets, leading to unreliable outputs. Beyond governance, technology infrastructure readiness is another key concern. AI applications require high-performance computing environments, scalable data processing pipelines and cloud-based architectures to function efficiently. The complex computational requirements and large data sets required to train AI models, which may include diverse data types, are straining data infrastructure and pushing the limits of traditional IT and data architectures.

AI risk management

As AI technology becomes more widely adopted, effective risk mitigation is essential for building confidence in this critical technology and ensuring responsible AI development and use. The adoption of broad-scale AI solutions necessitates robust machine learning operations (MLOps) to manage models efficiently and ensure scalability, version control and consistent performance. Robust data governance frameworks that emphasize data standardization, lineage tracking and compliance monitoring are critical to ensuring AI trustworthiness. Responsible AI principles, including accountability, transparency, security and reliability should be integrated throughout the software development and AI system lifecycle. As AI models evolve, real-world changes must be monitored, along with ongoing mitigation of partiality and risks to ensure model quality is maintained.

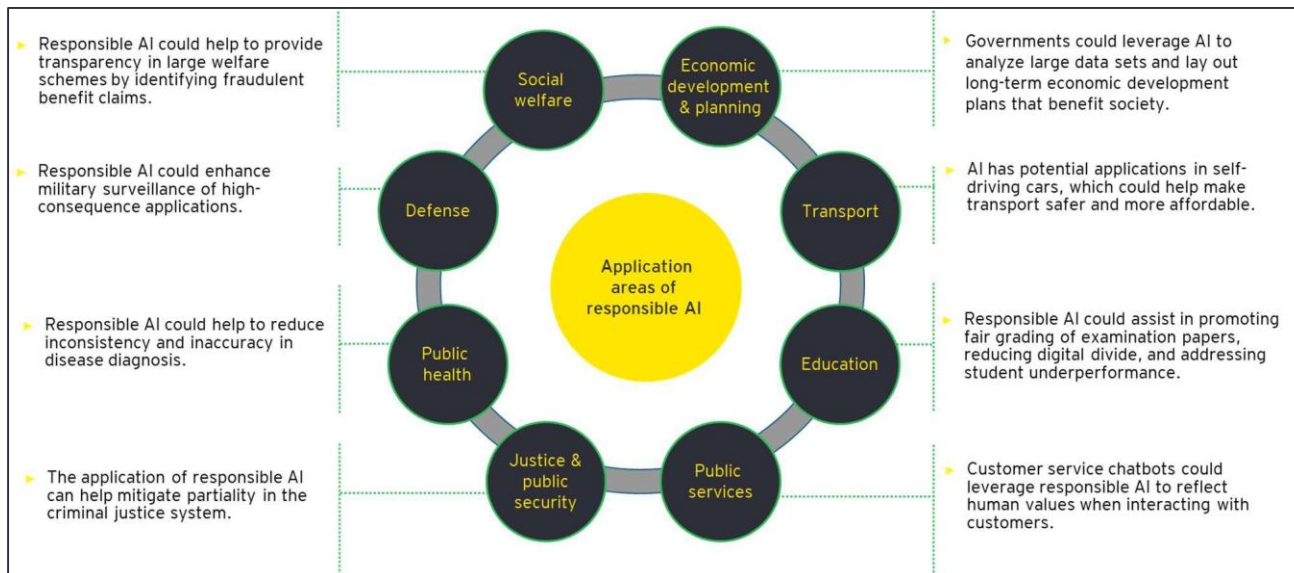


Figure 1. Responsible AI is expected to apply to a vast range of government domains in the future

Encouraging responsible AI development and deployment to support innovation

Supporting responsible AI, where risks are appropriately identified and managed, is essential to confident adoption and ensuring that our nation does not miss out on significant technological advancements that could enhance its global innovation leadership. Most US executives recognize AI's transformative potential and are actively investing in AI-driven initiatives. According to a [2024 EY survey](#), 61% of such leaders are also increasingly aware of the importance of responsible AI. In response to the growing AI landscape, governments worldwide are making substantial investments in AI and enhancing their R&D infrastructure by establishing AI-specific research hubs, advanced computing centers and centers of excellence. These efforts not only foster innovation by providing access to critical computing resources but also promote AI safety through the implementation of controlled sandbox environments. Ultimately, it is crucial to balance risks and rewards in a time- and cost-effective manner.

Cybersecurity throughout the lifecycle of AI system development and deployment

AI cybersecurity challenges and new attack vectors

AI systems introduce new cybersecurity threats, which are not as well understood as traditional threats. Both the dynamic and closed nature of many AI systems adds complexity relative to traditional technology systems, which often have clear and transparent decision-making paths with predictable behavior based on defined rules and protocols. This additional complexity makes traditional security assurance techniques, such as "point in time" security, less effective for AI systems. According to a 2024 Gartner study, almost 30% of enterprises deploying AI have already experienced an AI-related security breach. Examples of new attack vectors targeting AI systems include data poisoning, model inversion and adversarial attacks.

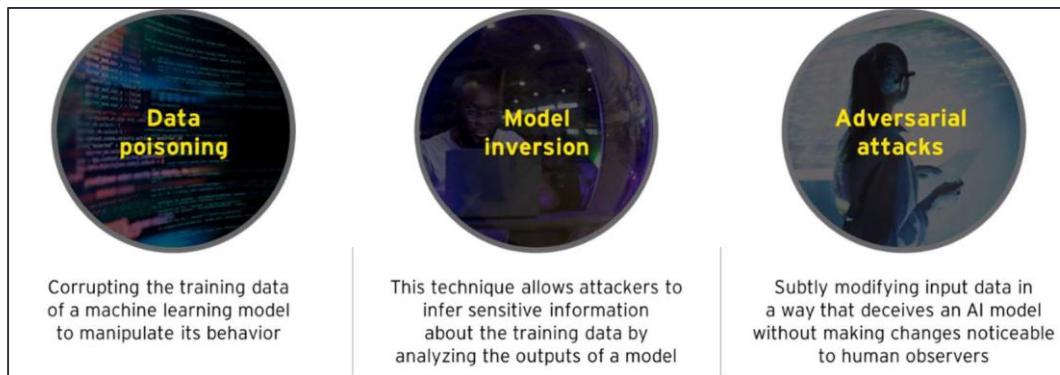


Figure 2. New cyber threats to AI systems

Threats throughout the AI lifecycle

Data security and privacy risks are present in each stage of the AI development and deployment lifecycle. The AI lifecycle is comprised of eight stages: use case initiation, data acquisition, data preparation, model experimentation, validation, deployment and production, monitoring and maintenance, and AI solution archiving.

At the early stages of the AI lifecycle (i.e., use case initiation, data acquisition, and data preparation), design and data risks present the greatest threats. These risks can involve an absence of data protection, data quality issues or a lack of robustness. When left unaddressed, these risks make AI systems vulnerable to cyber attacks that can compromise data security and deteriorate model performance.

As the AI lifecycle progresses to the model experimentation and validation stages, algorithmic risks emerge. These can come from the model exacerbating weaknesses in the data by creating a "feedback loop," poor scalability/performance, and selection of a "black box" model without explainability. Unresolved algorithmic risks often result in models that are too opaque for traditional assurance techniques to be applied to model internals.

After the model has been developed, performance risks can plague the deployment and production and monitoring and maintenance stages of the AI lifecycle. At this point, it is important to track model drift, as well as ensure accountability of the AI system. A lack of proper performance maintenance, such as AI-specific red teaming, creates challenges for incident response teams who can use performance data to quickly react to and mitigate potential cyber attacks. For these reasons, it is vital to incorporate cybersecurity support to mitigate risks that arise throughout the AI lifecycle.

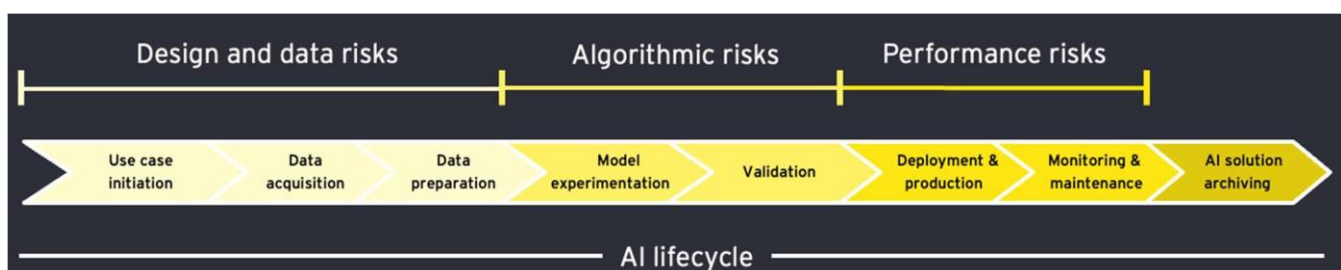


Figure 3. Risks throughout the AI development and deployment lifecycle can create cybersecurity vulnerabilities

AI security best practices

AI security must account for evolving attack vectors, require regular security assessments, resilient development practices, and real-time monitoring. Regular security testing, including adversarial evaluations and red teaming exercises, help organizations identify vulnerabilities (e.g., data poisoning, model inversion and prompt injection attacks) before they can be exploited. Secure development environments with built-in code validation, dependency management and vulnerability detection are key to strengthening AI system resilience. Ensuring data integrity through strict access controls, encryption and provenance tracking protects against unauthorized manipulation and supply chain risks.

Additionally continuous monitoring and automated detection capabilities enable organizations to proactively identify and mitigate emerging threats. AI-driven threat modeling and attack simulations further enhance attack preparedness, ensuring cybersecurity teams can adapt to new and evolving risks. As AI adoption expands, greater attention to proactive security measures, transparent risk assessments and scalable monitoring strategies will help organizations maintain trust and operational integrity while fostering responsible innovation.

Explainability of AI model outputs

What is AI explainability and when is it relevant?

The National Institute of Standards and Technology (NIST) has led efforts to support reliable and trustworthy use of AI since the 2019 executive order *Maintaining American Leadership in Artificial Intelligence*. A subsequent NIST study noted that explainability is "the mechanistic description of how a system made a specific prediction." That is, explainable AI algorithms seek to bring transparency to how a model makes decisions, thus removing the idea of a "black box" algorithm.

There are two main levels of explainability:

- 1) Global explanation, which refers to the ability to explain how a model makes predictions in general
- 2) Local explanation, which focuses on explaining how the model arrives at specific predictions for a given set of inputs

In addition to the two levels of explainability, there are also six different types of explainability depending on the requirements of the model and the business-specific considerations (as outlined in the figure below).



Figure 4. Types of AI explainability

Different models can require different levels and types of explainability. Explainability needs also depend on business specific considerations. For example, it is crucial for a health care provider to understand why AI has flagged a potential medical diagnosis, but it is less important for AI to be able to indicate why an email was flagged as spam. In general, AI explainability is necessary when transparency around the modeling design and outputs is necessary to establish model trust, confidence and fairness among its users.

Benefits of AI explainability

Explainability ensures that the AI model is suitable for its intended purpose and is protected against partiality before being deployed. This transparency builds trust and confidence and facilitates fairness. The benefits of AI explainability are numerous. It allows for the quick detection of errors and inconsistencies in AI models, enabling timely corrections. Furthermore, understanding the factors affecting AI model outputs allows for targeted optimizations, resulting in improved performance and efficiency. Insights into the relationships between input features and model outputs/predictions aids in making informed business decisions. Trust built by providing insights into how AI models make decisions leads to increased adoption of AI systems.

AI explainability challenges

While AI explainability models provide valuable insights into AI model behavior, they also pose some challenges, such as:

- Explainability - performance trade-off: AI models are often designed with the goal of achieving high accuracy or performance on a given task. However, as models become more complex and powerful, they tend to achieve higher accuracy but become less transparent and explainable. Highly complex models (e.g., deep neural networks) often provide superior accuracy but are inherently less interpretable, and simplifying the model for better explainability may result in a trade-off with accuracy.
- Data privacy and security concerns: Explainability techniques that rely on accessing and analyzing the model's internal information or training data can potentially raise privacy and security concerns. Accessing sensitive data or proprietary model details may introduce risks if not handled properly.
- Sensitivity to adversarial attacks: Some explainability techniques may be vulnerable to adversarial attacks, where malicious actors manipulate input features to mislead or deceive the explanations.
- Interpretation challenges to non-experts: Although AI explainability techniques aim to make an AI model understandable, some users may not have the necessary background knowledge to fully understand the explanations provided.

Implementing AI explainability

NIST has introduced guidance for determining whether an explanation adequately meets user needs. According to this guidance, an AI system should provide evidence and reasoning for its outcomes, offer understandable explanations that meet the needs of diverse users, accurately reflect its output generation process, and operate within its designed conditions while maintaining sufficient confidence in its outputs.

To achieve explainability based on this guidance, various eXplainable AI (XAI) algorithms have been developed and utilized, often falling into two broad categories:

- 1) Self-interpretable models, which refer to AI/ML models that are inherently designed to be easily understood and interpreted by humans, with simple and straightforward decision-making processes. Examples of self-interpretable models include linear regression, logistic regression, Generalized Linear Model (GLM), Generalized Additive Model (GAM), decision trees, decision rules, and Naïve Bayes Classifier.
- 2) Post-hoc explanations, which are explanations generated after the AI model has made a prediction. These post-hoc explanation algorithms are applied to existing complex AI models to shed light on their decision-making processes. Post-hoc explanations can be global, explaining how a model makes

predictions in general, or local, explaining how the model arrives at specific predictions for a given set of inputs. Instances of post-hoc explanations include example-based explanations, Local Interpretable Model-agnostic Explanations (LIME), SHapley Additive exPlanations (SHAP), Individual Conditional Expectation (ICE), Accumulated Local Effects (ALE), Partial Dependency Plot (PDP), and Friedman's H-statistic.

Each explainability model has unique advantages and disadvantages, including trade-offs between accuracy and interpretability (as discussed above), so selection of an explainability approach depends on the specific use case, stakeholders and explanation requirements.

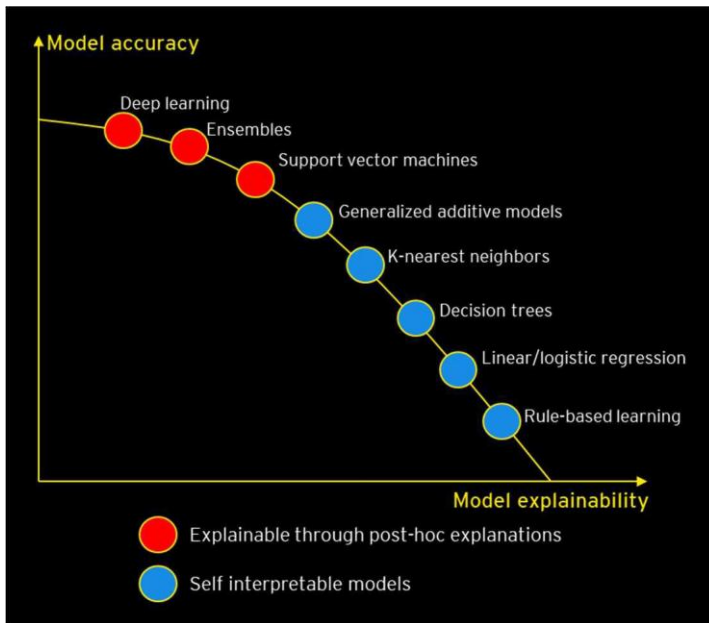
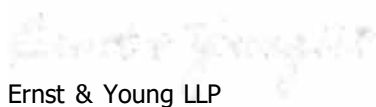


Figure 5. Explainability vs. accuracy for various AI/ML models

We commend OSTP and the NITRD National Coordination Office for seeking feedback on the development of an AI action plan. As we have described, identifying and managing risk through effective AI governance will support confidence and widespread adoption of AI in the US. Embracing responsibly AI innovation will allow the US to realize the enormous potential AI holds for people and businesses alike. We look forward to our continued engagement with your office as these important policies are developed.

If you have further questions, please contact John Hallmark in the EY Office of Public Policy

Yours sincerely,



Ernst & Young LLP